EDA On Application Data

Key Insights from Categorical Features vs TARGET (Defaulters %)

Employment Status:
- **Unemployed (36%)** show the **highest default rate** among all categories.
- **Businessman (0%)** and **Students (0%)** have the **lowest risk** of default.

Education Level:
- Clients with **Lower Secondary Education (11%)** have a **higher chance of default**.
- Those with an **Academic Degree (2%)** are the most reliable.

Gender:
- **Males (10%)** default more than **Females (7%)**.

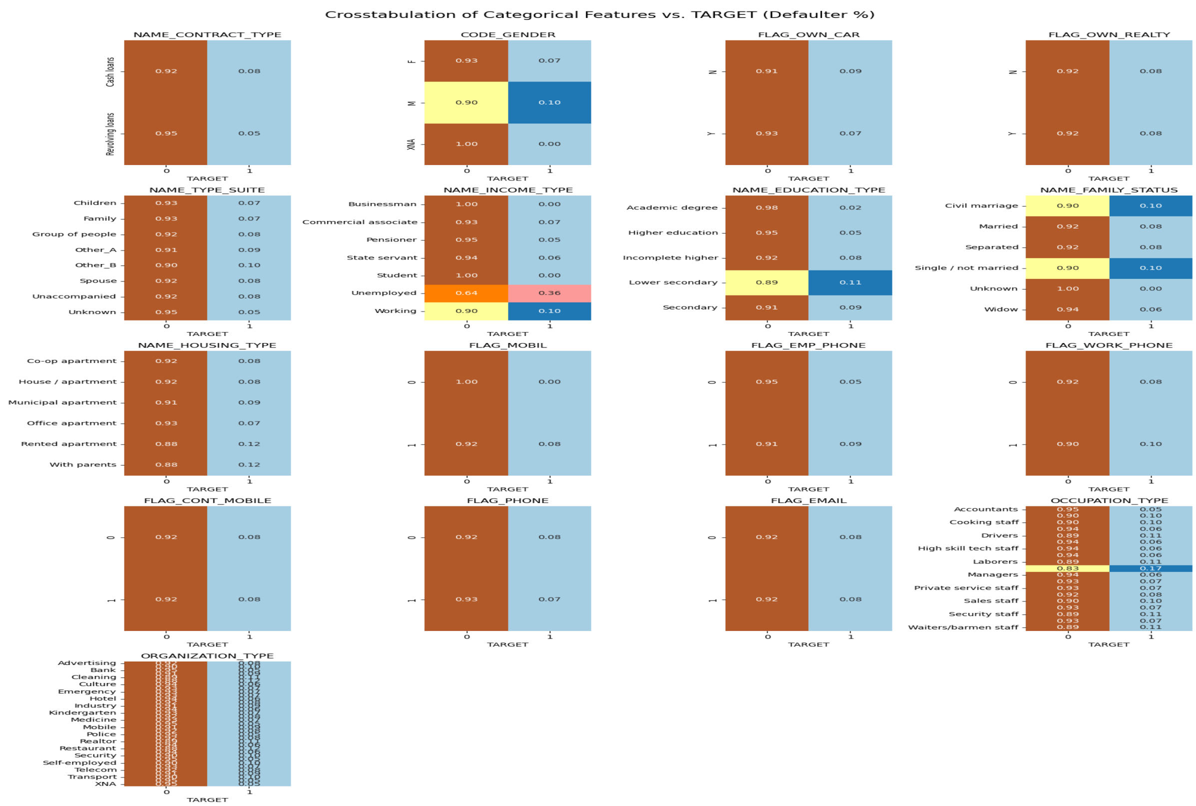
Loan Type:
- **Cash Loans (8%)** have a higher default rate than **Revolving Loans (5%)**.

Housing Type:
- People living **Rented (12%)** or **With Parents (12%)** show higher defaults.
- **House/Apartment owners (8%)** are more stable borrowers.

Family Status:
- **Single or Civil Marriage clients (10%)** have higher default rates than **Married (8%)**.

Occupation Type:
- **Low-skill Laborers (17%)** show the **highest default** among occupations.
- **Accountants (5%)** and **High-skill tech staff (6%)** are more creditworthy.

General Observation:
- Most categorical groups cluster around **7–9% default rate**.
- Employment, Education, income type, and housing situation are the **strongest predictors** of loan default likelihood.


Crosstabulation of Categorical Features vs. TARGET (Defaulter %)
NAME_CONTRACT_TYPE
Cash loans
Revolving loans
0.92
0.08
0.95
0.05
CODE_GENDER
F
M
XNA
0.93
0.07
0.90
0.10
1.00
0.00
FLAG_OWN_CAR
N
Y
0.91
0.09
0.93
0.07
FLAG_OWN_REALTY
N
Y
0.92
0.08
0.92
0.08
NAME_TYPE_SUITE
Children
Family
Group of people
Other_A
Other_B
Spouse
Unaccompanied
Unknown
NAME_INCOME_TYPE
Businessman
Commercial associate
Pensioner
State servant
Student
Unemployed
Working
NAME_EDUCATION_TYPE
Academic degree
Higher education
Incomplete higher
Lower secondary
Secondary
NAME_FAMILY_STATUS
Civil marriage
Married
Separated
Single / not married
Unknown
Widow
NAME_HOUSING_TYPE
Co-op apartment
House / apartment
Municipal apartment
Office apartment
Rented apartment
With parents
FLAG_MOBIL
FLAG_EMP_PHONE
FLAG_WORK_PHONE
FLAG_CONT_MOBILE
FLAG_PHONE
FLAG_EMAIL
OCCUPATION_TYPE
Accountants
Cooking staff
Drivers
High skill tech staff
Laborers
Managers
Private service staff
Sales staff
Security staff
Waiters/barmen staff
ORGANIZATION_TYPE
Advertising
Bank
Cleaning
Culture
Emergency
Hotel
Industry
Kindergarten
Medicine
Mobile
Police
Realtor
Restaurant
Security
Self-employed
Telecom
Transport
XNA
TARGET
0
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